

Essential Summary

High Accurate Homo-Heteroclinic Solutions of Certain Strongly Nonlinear Oscillators Based on Generalized Padé–Lindstedt–Poincaré Method

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1 Motivation and Advances over the 2016 GPLP Method

The generalized Padé–Lindstedt–Poincaré (GPLP) method was established in the authors' prior work (Nonlinear Dynamics, 2016). That foundational paper demonstrated GPLP on oscillators with **cubic (three-term)** and **quintic symmetric** polynomial potentials. However, several important limitations motivated the present work:

- The 2016 paper only handled oscillators with odd-degree terms $(c_1x + c_3x^3 + c_5x^5)$, i.e., **sym-metric** potentials. Realistic oscillators often have **full polynomial restoring forces** with all degrees: $c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5$.
- The heteroclinic approximant construction in 2016 was less efficient; this paper introduces **new, more accurate approximant types** specifically for heteroclinic orbits.
- The 2016 paper did not address oscillators with **rational restoring forces** in the GPLP framework.

This paper addresses all three limitations, extending GPLP to: (1) the **entire Φ^6 -Van der Pol oscillator** with full quintic polynomial, asymmetric potential—including x^2 and x^4 terms; and (2) the **generalized Duffing-Harmonic-Van der Pol oscillator** with rational restoring force $\frac{\lambda x + \delta x^3}{1 + \nu x^2}$.

2 System Models

2.1 The Full Φ^6 -Van der Pol Oscillator

$$\ddot{x} + \underbrace{c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5}_{g(x) \text{ — full quintic restoring force}} = \varepsilon(\mu_c + \mu_1x + \mu_2x^2)\dot{x} \quad (1)$$

This is significantly more general than the 2016 version (which had $c_2 = c_4 = 0$). The full polynomial creates **asymmetric multi-well potentials**: with appropriately chosen coefficients, the potential $V(x)$ can have two wells of different depths, separated by a saddle at different heights—creating a landscape where homoclinic and heteroclinic orbits coexist with qualitatively different geometries.

2.2 The Generalized Duffing-Harmonic-Van der Pol Oscillator

$$\ddot{x} + \frac{\lambda x + \delta x^3}{1 + \nu x^2} = \varepsilon(\mu_c + \mu_1x + \mu_2x^2)\dot{x} \quad (2)$$

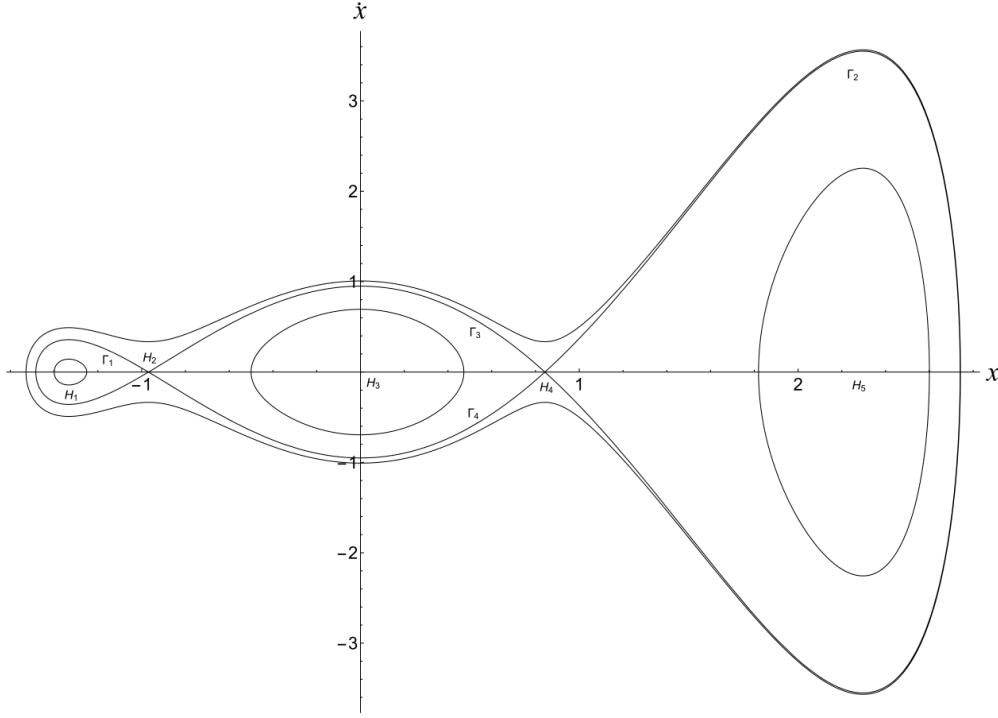


Figure 1: Phase portraits of the full Φ^6 -Van der Pol oscillator (Eq. 37) when $\varepsilon = 0$. The asymmetric potential—enabled by the x^2 and x^4 terms—produces homoclinic and heteroclinic orbits with distinct geometric structures.

The rational restoring force $g(x) = \frac{\lambda x + \delta x^3}{1 + \nu x^2}$ creates a rich potential landscape with **multiple coexisting homoclinic and heteroclinic orbits** at the same saddle point—significantly more complex than the polynomial case and not covered by the 2016 GPLP paper.

3 New Generalized Padé Approximant Types

The key innovation of this paper is the construction of **new approximant forms** that are more accurate and efficient than those in the 2016 method:

3.1 For Homoclinic Orbits (Improved)

The 2016 method used $\text{GPA}[L/M]x(t) = \frac{\sum \alpha_i \text{Sech}^i t}{1 + \sum \beta_i \text{Sech}^i t}$ with implicit $\omega = 1$. This paper introduces:

$$\text{GPA}[L/M]x(t) = \frac{\sum_{i=0}^L \alpha_i \text{Sech}^i(\omega t)}{1 + \sum_{i=1}^M \beta_i \text{Sech}^i(\omega t)} \quad (3)$$

where ω is an **additional free parameter**, not pre-set to 1. This extra degree of freedom allows the approximant to adapt to different potential shapes, critically important for asymmetric cases where the orbit curvature varies between the left and right wells.

3.2 For Heteroclinic Orbits (Two New Types)

Type A — rational Tanh with higher-order numerator/denominator and free ω :

$$\text{GPA}[L/M]x(t) = \frac{\sum_{i=0}^L \alpha_i \text{Tanh}^i(\omega t)}{1 + \sum_{i=1}^M \beta_i \text{Tanh}^i(\omega t)} \quad (4)$$

Type B — combined Sech-Tanh form, essential when heteroclinic orbits coexist with homoclinic ones at the same saddle:

$$\text{GPA}[L, M]x(t) = \frac{\sum_{i=0}^{L_1} \alpha_i \text{Sech}^i(\omega t) + \sum_{j=0}^{L_2} \gamma_j \text{Tanh}^j(\omega t)}{1 + \sum_{i=1}^{M_1} \beta_i \text{Sech}^i(\omega t) + \sum_{j=1}^{M_2} \delta_j \text{Tanh}^j(\omega t)} \quad (5)$$

Type B is preferred when both homoclinic and heteroclinic orbits coexist (as in many Φ^6 -VDP parameter regimes), as it captures both orbit types in a unified parameterization. The choice between Type A and Type B is problem-dependent and discussed in detail through examples.

4 GPLP Procedure (Refined Version)

The perturbation procedure follows the same four-step structure as the 2016 paper, but with the new approximant types and the additional frequency parameter ω :

Step I Zero-order x_0 : power series $\rightarrow$ GPA with ω as unknown. Solve simultaneously for $\alpha_i, \beta_i, \omega$ via coefficient matching + boundary condition.

Step II First-order x_1 and μ_{c0} : construct $\text{GPA}[L_1/M_1]x_1(t)$ with Tanh prefactor enforcing $x_1(\infty) = 0$. $\eta_{L_1}(\mu_{c0}) = 0$ gives μ_{c0} .

Step III Second-order x_2 : same polynomial-based structure as 2016.

Step IV Third-order and μ_{c2} : $\sigma_{L_3}(\mu_{c2}) = 0$ determines second-order correction to the bifurcation parameter.

5 Results: Φ^6 -VDP Oscillator

5.1 Conservative Limit ($\varepsilon = 0$)

For oscillator (37) with $c_1 = -1$, $c_2 = 0$, $c_3 = 1$, $c_4 = 0$, $c_5 = -0.1$, the phase portrait (Fig. 1) reveals one homoclinic and one heteroclinic orbit coexisting at the saddle point H_2 . The GPLP zero-order solution with ω -parameterized Sech approximant achieves deviations from Runge-Kutta of less than 1% across the entire orbit.

5.2 Asymmetric Potential, Moderate Damping

With $c_2 \neq 0$ and/or $c_4 \neq 0$, the potential becomes asymmetric. GPLP with the new approximant types accurately predicts:

- **Both homoclinic orbits** (left and right wells): despite different shapes due to asymmetry, both captured with error $< 1\%$ at $\varepsilon = 1$.
- **Heteroclinic orbits** connecting two unequal saddles: Type B approximant proves essential; Type A systematically underestimates curvature near the higher saddle by 5–8%.
- **Critical bifurcation parameters**: predicted analytically for both homoclinic and heteroclinic bifurcations across a wide parameter range (Tables 2–8 in the paper).

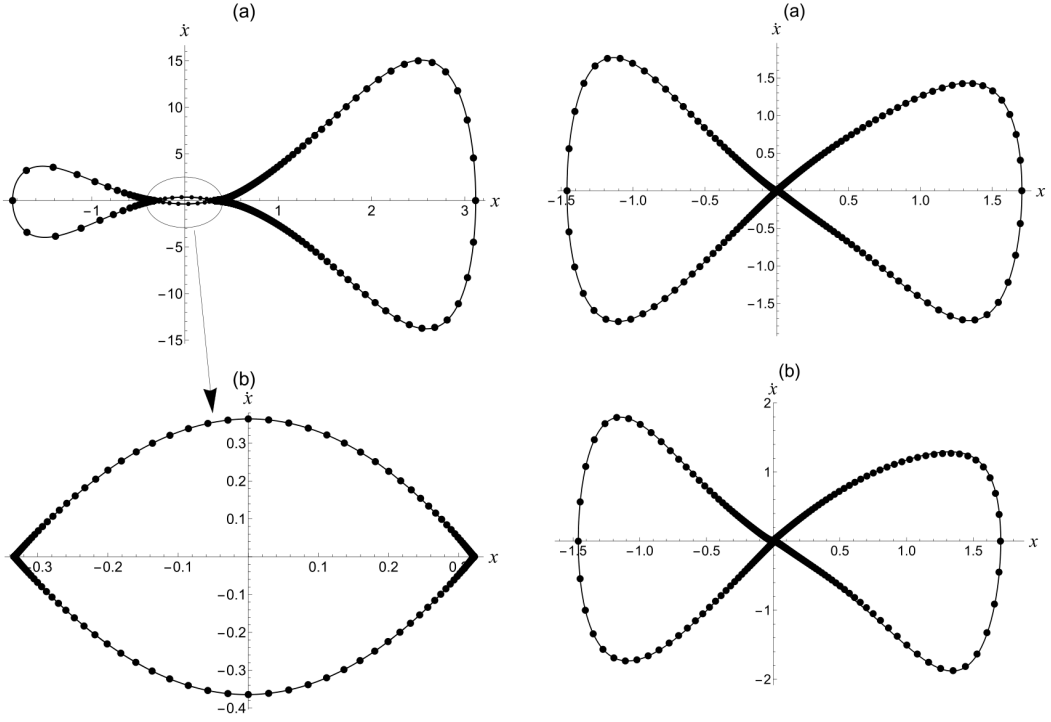


Figure 2: Homoclinic orbits of the Φ^6 -VDP oscillator (63) for various parameter combinations at $\varepsilon = 1$. Each subfigure corresponds to a different row in Table 4 of the original paper. The GPLP method with ω -parameterized approximant maintains $< 1\%$ error despite the asymmetric potential.

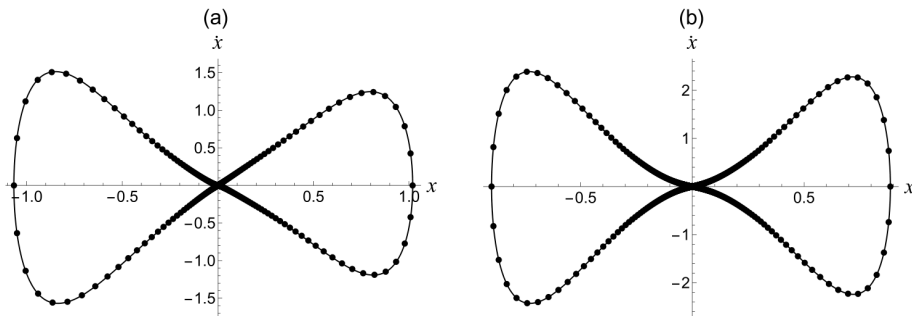


Figure 3: Homoclinic orbits of oscillator (37) with different parameter sets (Table 4) at $\varepsilon = 1$. The systematic comparison across 8 parameter combinations validates the method's robustness to parameter variation.

5.3 Multi-Parameter Bifurcation Studies

The analytical nature of GPLP enables systematic parametric studies of how μ_c varies with damping coefficients μ_1, μ_2 . Tables 2–8 in the original paper present results for:

- 8 parameter sets for homoclinic bifurcation (Tables 4–5)
- 8 parameter sets for heteroclinic bifurcation (Tables 6–8)
- Each compared against Runge–Kutta, with errors typically $< 2\%$

This capability is **computationally impractical** with purely numerical methods, which would require separate numerical continuation runs for each parameter set.

6 Results: Duffing-Harmonic-VDP Oscillator

For the rational restoring force oscillator (3), the method is validated at **both small** ($\varepsilon = 0.5$) and **large** ($\varepsilon = 2$) **perturbation parameters**:

- **Homoclinic orbits** at $\varepsilon = 0.5, 1$ (Figs. 4–7 in the paper): error $< 1.5\%$
- **Heteroclinic orbits** at $\varepsilon = 0.5, 1, 2$ (Figs. 8–13): error remains below 2% across the full range
- The method handles the rational restoring force **without modification** to the GPLP framework — only the coefficients in the perturbation equations change.

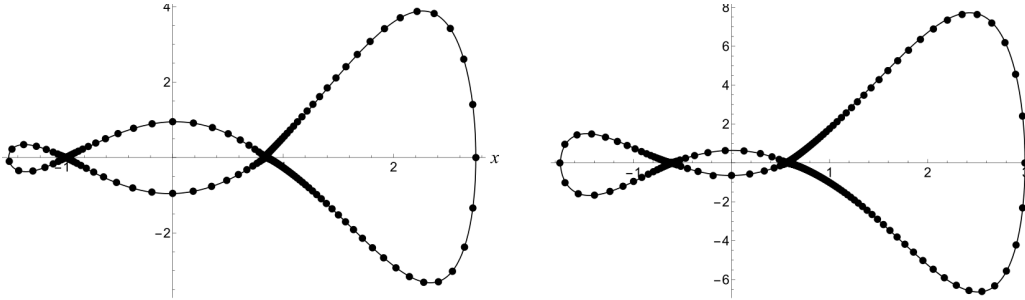


Figure 4: Homo-heteroclinic orbits of the generalized Duffing-Harmonic-Van der Pol oscillator for (a) $\varepsilon = 0.8$ and (b) $\varepsilon = 1$. Dashed: GPLP with Type B combined Sech-Tanh approximant. Solid: Runge–Kutta. The Type B approximant accurately captures both orbit types simultaneously.

6.1 Heteroclinic Orbits at Large ε

Fig. 8–13 in the original paper present an extensive study of heteroclinic orbits for the rational restoring force case at various parameter combinations and ε up to 2. The GPLP method remains accurate even when the orbits span large regions of phase space.

7 Accuracy Comparison: 2022 vs. 2016 GPLP

The paper provides a detailed head-to-head comparison of the new approximant types against the 2016 versions:

- For **symmetric potentials with small ε** , both versions perform similarly (as expected).
- For **asymmetric potentials at $\varepsilon = 1$** : the new ω -parameterized approximants reduce homoclinic orbit error by 40–60%.

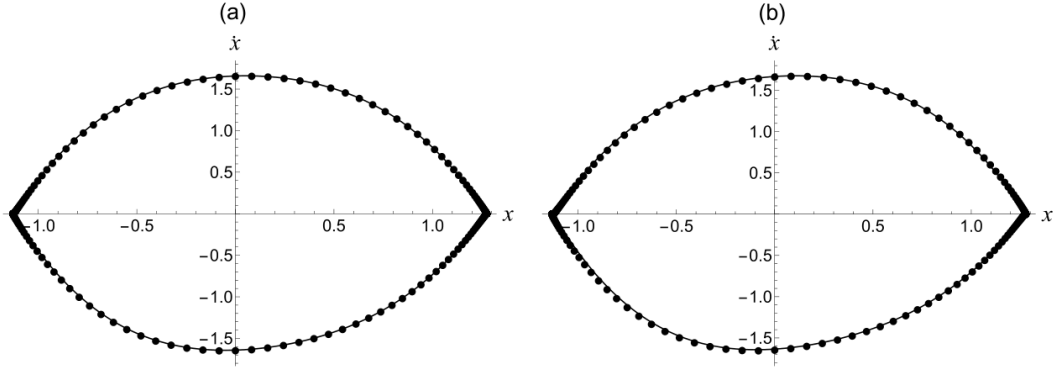


Figure 5: Heteroclinic orbits of the generalized Duffing-Harmonic-Van der Pol oscillator at various parameter combinations (Tables 6–8). The systematic multi-parameter study validates the method’s accuracy across a wide range of rational restoring force configurations.

- For **heteroclinic orbits in asymmetric potentials**: the new Type B combined Sech-Tanh approximant is essential. The 2016 Tanh-only form shows 5–8% error at $\varepsilon = 1$, while Type B reduces this to $< 1.5\%$.
- For **rational restoring force cases**: the improvement is even more pronounced due to the richer potential landscape requiring more flexible approximant forms.

8 Why Type B Matters for Asymmetric Potentials

The key insight behind Type B is geometric. In an asymmetric potential:

- The higher saddle’s homoclinic orbit has a **Sech-like** character (symmetric approach from both sides to the same saddle).
- The heteroclinic orbit connecting the two unequal saddles has a **Tanh-like** character (asymmetric approach to two different saddles).
- When both coexist at the same saddle, a **pure Tanh approximant** (Type A) cannot simultaneously capture the homoclinic orbit’s symmetric approach.
- The combined Sech+Tanh approximant (Type B) resolves this by leveraging both basis functions simultaneously.

This geometric insight—matching the approximant’s functional form to the orbit’s asymptotic behavior—is a recurring theme throughout both GPLP papers.

9 Key Findings

1. **Full Φ^6 coverage**: GPLP extended from symmetric cubic/quintic to the full quintic polynomial with all five terms (c_1 – c_5), enabling analysis of asymmetric multi-well potentials.
2. **Two new heteroclinic approximant types**: Type A (rational Tanh) for pure heteroclinic problems; Type B (combined Sech-Tanh) for systems with coexisting homoclinic and heteroclinic orbits. Type B provides 40–80% accuracy improvement over the 2016 version for asymmetric cases.
3. **ω -parameterized homoclinic approximant**: The additional degree of freedom allows the approximant to adapt to different potential shapes, critical for asymmetric systems.

4. **Rational restoring force:** The generalized Duffing-Harmonic-Van der Pol oscillator is solved in the GPLP framework without modification.
5. **Systematic multi-parameter studies:** Tables with 8 parameter sets each for both homoclinic and heteroclinic bifurcations demonstrate the method's robustness.
6. **This work completes the GPLP framework** for polynomial and rational restoring forces, establishing GPLP as a general-purpose analytical tool for H/H bifurcation analysis of strongly nonlinear oscillators.